

Brain

The brain contains sensors that continually monitor the levels of substances, such as sugar and water, in the blood. When action is needed to regulate the levels, the brain sends signals—either nerve impulses or hormones—to parts of the body that are able to fix this, such as the kidneys.

Basking in the sun

Reptiles, such as the marine iguana, rely on the external environment to regulate their temperature and use the sun's heat to warm themselves up. However, mammals and birds generate body heat to keep their temperature constantly warm.

**Dealing with salt**

A diet of seaweed and an ocean life is high in salt. But too much salt damages cells, drawing water from them and making them dehydrated. Marine iguanas are able to stop the levels of salt from getting too high by removing the excess. Glands in the nose concentrate the salt into mucus and then an explosive sneeze scatters the salty spray.

Marine Iguana

Found only in the Galápagos Islands, marine iguanas live an unusual life—diving for food in cold oceans, and then basking on rocks to warm up in the sun's rays. Homeostasis is carried out in each iguana's body by a complex variety of organs and glands. Glands are organs that produce chemical substances called hormones—tiny messengers that travel through the bloodstream to signal the relevant organ to take action.

Liver

The largest organ of the body is also one of the busiest. As well as regulating blood sugar levels, the liver has the job of neutralizing any poisons that enter the body. It passes waste back into the blood to be picked up by the kidneys.

